

Lecture 6: Multiple linear regression

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Materials in this lecture notes come from two lecture notes written by Jean Kyung Kim (Inha Univ.) and Pratheepa Jeganathan (Stanford Univ.).



Multiple linear regression

Outline

- ▶ Specifying the model.
- ▶ Fitting the model: least squares.
- ▶ Interpretation of the coefficients.

Model

- ▶ Given a sample $Y_i, X_{i1}, X_{i2}, \dots, X_{ip}, i = 1, 2, \dots, n$, the multiple linear regression model is



$$y_i = \underbrace{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}}_{\mathbb{E}[Y|X_1, \dots, X_p]} + \epsilon_i,$$

where $i = 1, 2, \dots, n$.

- ▶ y_i : i -th value of the response variable Y .
- ▶ x_{ij} : j -th predictor variable for the i -th unit.
- ▶ Assumptions:
 - ▶ Errors ϵ are assumed independent $N(0, \sigma^2)$, as in simple linear regression.
- ▶ Coefficients are called *partial* regression coefficients because they “allow” for the effect of other variables.

Example (Prostate data)

- For more information on the **Gleason score**.

Variable	Notation	Description
lpsa	Y	(log) Prostate Specific Antigen
lcavol	X_1	Cancer Volume
lweight	X_2	Weight
age	X_3	Patient age
lbph	X_4	(log) Venning Prostatic Hyperplasia
svi	X_5	Seminal Vesicle Invasion
lcp	X_6	(log) Capsular Penetration
gleason		Gleason score
pgg45	X_7	Percent of Gleason score 4 or 5
train		Label for test or training split

- We assume a linear model relating Y and seven explanatory variables.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_5 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \epsilon.$$

Example

- ▶ This data set is from the book [The Elements of Statistical Learning](#).

```
library(xtable)
prostate = read.table(file='./prostate.txt', header=T)
head(prostate)[,1:6]
```

##		lcavol	lweight	age	lbph	svi	lcp
## 1		-0.5798185	2.769459	50	-1.386294	0	-1.386294
## 2		-0.9942523	3.319626	58	-1.386294	0	-1.386294
## 3		-0.5108256	2.691243	74	-1.386294	0	-1.386294
## 4		-1.2039728	3.282789	58	-1.386294	0	-1.386294
## 5		0.7514161	3.432373	62	-1.386294	0	-1.386294
## 6		-1.0498221	3.228826	50	-1.386294	0	-1.386294

Parameter estimation

- ▶ Just as in simple linear regression, model is fit by minimizing

$$\text{SSE}(\beta_0, \dots, \beta_p) = \sum_{i=1}^n \left(Y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j X_{ij} \right) \right)^2$$

- ▶ By a direct application of calculus, it can be shown that the least squares estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_{0p}$ are given by the solution of a system of linear equations known as *normal equation*.
 - ▶ $\hat{\beta}_0$ is intercept (may or may not be $\bar{Y}$.)
 - ▶ $\hat{\beta}_j$ is the estimate of the partial regression coefficient of the predictor X_j .

(conti.)

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$* LSE \quad \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 = f(\beta_0, \beta_1, \dots, \beta_p)$$

$$\frac{\partial f}{\partial \beta_0} = \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j) (-1) = 0$$

$$\frac{\partial f}{\partial \beta_1} = \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j) (-x_{i1}) = 0$$

$$\vdots$$

$$\frac{\partial f}{\partial \beta_p} = \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j) (-x_{ip}) = 0$$

normal equation

$$\begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$$

$$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$$

$$\hookrightarrow \sum_{i=1}^n e_i = 0, \quad \sum_{i=1}^n e_i x_{i1} = 0, \quad \dots, \quad \sum_{i=1}^n e_i x_{ip} = 0 \rightarrow$$

$$e_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip}$$

$$1_n \perp e$$

$$x_1 \perp e$$

$$\vdots$$

$$x_p \perp e$$

Parameter estimation for the example

```
options(xtable.comment = FALSE)
prostate.lm = lm(lpsa ~ lcavol + lweight +
  age + lbph + svi + lcp + pgg45,
  data = prostate)
print(xtable(prostate.lm, digits =3), scalebox=0.8)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.494	0.874	0.566	0.573
lcavol	0.570	0.086	6.634	0.000
lweight	0.614	0.198	3.096	0.003
age	-0.021	0.011	-1.905	0.060
lbph	0.097	0.058	1.691	0.094
svi	0.752	0.238	3.159	0.002
lcp	-0.105	0.089	-1.175	0.243
pgg45	0.005	0.003	1.573	0.119

Estimating σ^2

- ▶ As in simple regression

$$\widehat{\sigma}^2 = \frac{SSE}{n - p - 1}.$$

- ▶ We can show that

$$\frac{(n - p - 1)\widehat{\sigma}^2}{\sigma^2} \sim \chi_{n-p-1}^2$$

- ▶ Why χ_{n-p-1}^2 ?
 - ▶ Typically, the degrees of freedom in the estimate of σ^2 is $n - \text{\#number of parameters in regression function}$.

Estimating σ^2 for the example

- The degrees of freedom

```
prostate.lm$df.resid
```

```
## [1] 89
```

- Using the formula

```
sigma.hat = sqrt(sum(resid(prostate.lm)^2)  
/df.residual(prostate.lm))  
round(sigma.hat, digits = 3)
```

```
## [1] 0.696
```

- OR

```
round(sqrt(deviance(prostate.lm)/df.residual(prostate.lm)), digits = 3)
```

```
## [1] 0.696
```

Interpretation of regression coefficients β_j 's

- ▶ Take $\beta_1 = \beta_{\text{l cavol}}$ for example. This is the amount the average `lpsa` rating increases for one “unit” of increase in `l cavol`, keeping everything else constant.
- ▶ We refer to this as the effect of `l cavol` *allowing for* or *controlling for* the other variables.

Example (interpretation of β_j)

- For example, let's take the 10th case in our data and change `lcavol` by 1 unit.

```
options(xtable.comment = FALSE)
case10 = prostate[10,]
print(xtable(case10), scalebox=0.8)
```

	lcavol	lweight	age	lbph	svi	lcp	gleason	pgg45	lpsa	train
10	0.22	3.24	63	-1.39	0	-1.39	6	0	1.05	FALSE

```
case10.temp = case10
case10.temp['lcavol'] =
  case10.temp['lcavol'] + 1
```

Example (interpretation of β_j)

```
Yhat = predict(prostate.lm, rbind(case10, case10.temp))
names(Yhat) = c("lcavol_10", "lcavol_10+1")
Yhat
```

```
##    lcavol_10 lcavol_10+1
##      1.307754      1.877300
```

- Our regression model says that this difference should be $\hat{\beta}_{1\text{cavol}}$.

```
c(Yhat[2]-Yhat[1], coef(prostate.lm)['lcavol'])
```

```
## lcavol_10+1      lcavol
##      0.569546      0.569546
```

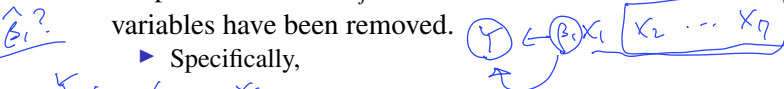
Partial regression coefficients

- ▶ The term *partial* refers to the fact that the coefficient β_j represent the partial effect of X_j on Y , i.e. after the effect of all other variables have been removed.

- ▶ Specifically,

$$Y_i - \sum_{l=1, l \neq j}^k X_{il}\beta_l = \beta_0 + \beta_j X_{ij} + \varepsilon_i.$$

- ▶ Let $e_{i,(j)}$ be the residuals from regressing Y onto all X 's except X_j , and let $X_{i,(j)}$ be the residuals from regressing X_j onto all X 's except X_j .
 - ▶ $e_{i,(j)} = Y - (\hat{b}_0 + \hat{b}_1 X_1 + \cdots + \hat{b}_{j-1} X_{j-1} + \hat{b}_{j+1} X_{j+1} + \cdots + \hat{b}_p X_p).$
 - ▶ $X_{i,(j)} = X_j - (\hat{a}_0 + \hat{a}_1 X_1 + \cdots + \hat{a}_{j-1} X_{j-1} + \hat{a}_{j+1} X_{j+1} + \cdots + \hat{a}_p X_p).$
- ▶ If we regress $e_{i,(j)}$ against $X_{i,(j)}$, the coefficient is *exactly* the same as in the original model.
 - ▶ $e_{i,(j)} = C_0 + \beta_j X_{i,(j)} + \epsilon_i.$



Example

- Let's verify this interpretation of regression coefficients.

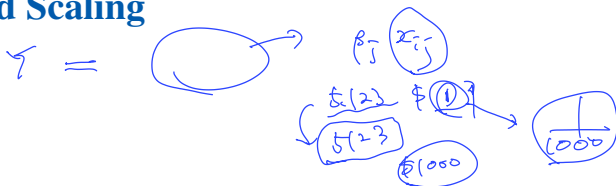
```
partial_resid_lcavol = resid(lm(lcavol ~ lweight +  
  age + lbph + svi +  
  lcp + pgg45, data=prostate))  
partial_resid_lpsa = resid(lm(lpsa ~ lweight +  
  age + lbph + svi +  
  lcp + pgg45, data=prostate))
```



```
print(xtable(summary(lm(partial_resid_lpsa ~
  partial_resid_lcavol))))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0000	0.0684	0.00	1.0000
partial_resid_lcavol	0.5695	0.0831	6.85	0.0000

Centering and Scaling



- ▶ Regression coefficient magnitude depends on the unit of measurements of the variable.
 - ▶ If regression coefficient of income is 5.123 when measured in dollars, but 5123 when measured in \$1000.
- ▶ To make regression coefficient unitless
 - ▶ Can *center* and *scale* the variables.

Centering and scaling for the intercept model

► Centering $X_j - \bar{x}_j$ and $Y - \bar{y}$.

► Scaling

► Unit-length scaling:

and

$$\tilde{X}_j = \frac{X_j - \bar{x}_j}{L_j}, L_j = \sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}, j = 1, 2, \dots, p.$$

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i$$

$$y_i - \bar{y} = \cancel{\beta_0} + \beta_1 (x_{i1} - \bar{x}_1) + \dots + \beta_p (x_{ip} - \bar{x}_p) + \varepsilon_i$$

(LSE)

$$\|z\|_2 = \sqrt{\sum_{i=1}^n z_i^2} = 1$$

$$\tilde{Z}_y = \frac{Y - \bar{y}}{L_y}, L_y = \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}$$

(conti.)

$$\frac{y_i}{s_y} = \beta_0 + \beta_1 \frac{x_{i1}}{s_{x_1}} + \beta_2 \frac{x_{i2}}{s_{x_2}} + \varepsilon_i$$

► Scaling

► Standardizing:

$$s_{zy} = 1$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$\tilde{Z}_y = \frac{Y - \bar{y}}{s_y}, s_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}$$

and

$$\tilde{y}_i = s_y \cdot \beta_0 + \beta_1 \frac{s_y}{s_{x_1}} x_{i1} + \beta_2 \frac{s_y}{s_{x_2}} x_{i2} + \varepsilon_i$$

$$\tilde{X}_j = \frac{X_j - \bar{x}_j}{s_j}, s_j = \sqrt{\frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}{n-1}}, j = 1, 2, \dots, p.$$

Centering and scaling for the intercept model

- ▶ Let $\hat{\theta}_j$ be the estimated regression coefficient for the standardized data.
 - ▶ The estimated regression coefficient for the original data are

$$\hat{\beta}_j = \frac{s_y}{s_j} \hat{\theta}_j, j = 1, 2, \dots, p,$$

and

$$\hat{\beta}_0 = \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{x}_j.$$

- ▶ θ_j measures the change in standardized units of Y corresponding to an increase of one standard deviation in X_j .

* Original model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i$$

* centered & scaled

$$\frac{Y_i - \bar{Y}}{s_y} = \theta_1 \frac{x_{i1} - \bar{x}_1}{s_1} + \theta_2 \frac{x_{i2} - \bar{x}_2}{s_2} + \dots + \theta_p \frac{x_{ip} - \bar{x}_p}{s_p} + \eta_i$$

$$\hookrightarrow Y_i = \underbrace{\bar{Y} - \frac{s_y}{s_1} \theta_1 \bar{x}_1 - \dots - \frac{s_y}{s_p} \theta_p \bar{x}_p}_{\beta_0} + \theta_1 \frac{s_y}{s_1} x_{i1} + \dots + \theta_p \frac{s_y}{s_p} x_{ip} + \eta_i$$

$$\beta_j = \theta_j \cdot \frac{s_y}{s_j}$$

Scaling for no-intercept model

- ▶ Centering has an effect of including an intercept in the model.
- ▶ Let

$$Y = \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

be the no-intercept model.

- ▶ Scaled variables are

$$\tilde{Z}_y = \frac{Y}{L_y}, L_y = \sqrt{\sum_{i=1}^n y_i^2},$$

and

$$\tilde{X}_j = \frac{X_j}{L_j}, L_j = \sqrt{\sum_{i=1}^n x_{ij}^2}, j = 1, 2, \cdots, p.$$

References for this lecture

- ▶ **HC** Chapter 4.1-4.6
- ▶ Lecture notes of [Jonathan Taylor](#) .